



The School of Systems and Technology (SST)

Complex Computing Porject

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Course :

Computer Networks Lab (V-9)

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Introduction:

This project demonstrates the implementation of an ISP Network using GNS3. The network consists of three routers connected in a triangle topology to provide redundancy and backup communication paths.

Static routing was configured between routers, and a backup route was implemented using Administrative Distance. The network was tested successfully using ping commands and failover simulation.

Objectives:

The objectives of this project are:

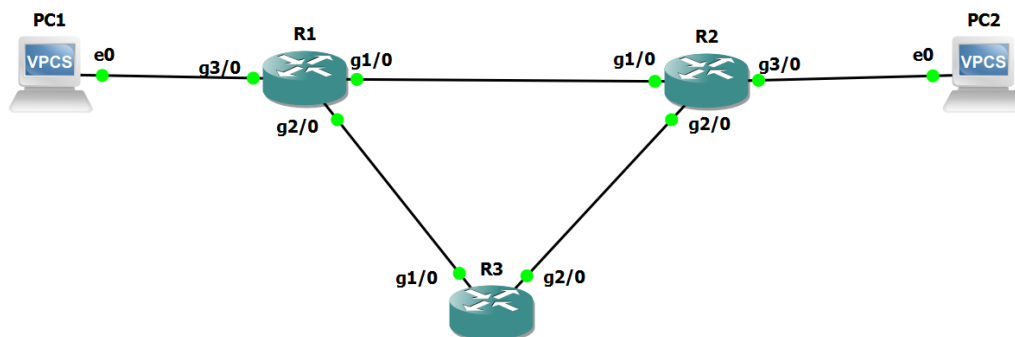
- Design ISP topology in GNS3
- Configure router interfaces and IP addressing
- Configure static routing
- Implement backup route using Administrative Distance
- Test connectivity between devices
- Configure SSH remote access
- Simulate failover mechanism

Network Topology:

The topology contains:

Devices	Quantity
Routers	3
PC	2

Topology Structure:



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IP Addressing Scheme:

Devices	Network	IP Address
R1 ↔ R2	10.0.12.0/30	10.0.12.1 / 10.0.12.2
R1 ↔ R3	10.0.13.0/30	10.0.13.1 / 10.0.13.2
R2 ↔ R3	10.0.23.0/30	10.0.23.2 / 10.0.23.3
PC1		192.168.1.10
PC2		192.168.2.10

Router Configuration Summary:

R1:

- Connected to R2 and R3
- Configured LAN gateway for PC1
- Configured primary and backup static routes
- Configured SSH access

R2:

- Connected to R1 and R3
- Configured gateway for PC2
- Configured return static route

R3:

- Configured as backup router
- Used for failover communication path

Static Routing and Administrative Distance:

Lower Administrative Distance is preferred by the router.

Normally traffic uses:

R1 → R2

If the primary link fails, traffic switches automatically to:

R1 → R3 → R2

This process is called failover.

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Connectivity Testing:

Ping Test Matrix

Source	Destination	Result
PC1	PC2	Success
PC1	R1	Success
PC1	R2	Success
PC1	R3	Success
PC2	PC1	Success
PC2	R1	Success
PC2	R3	Success
R1	R2	Success
R1	R3	Success
R2	R3	Success

Failover Testing

The primary interface between R1 and R2 was shut down to test failover.

Shutdown Command:

```
configure terminal
interface g1/0
shutdown
```

Result

After the primary link failure:

- backup route became active
- communication remained successful
- traffic automatically used alternate path

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Traffic path after failover:

R1 → R3 → R2

This confirmed successful redundancy implementation.

Conclusion:

The ISP Network project was implemented successfully in GNS3.

The project successfully demonstrated:

- Static routing
- Redundant topology
- Automatic failover
- End-to-end connectivity
- SSH remote management

The backup route maintained communication during primary link failure, improving network reliability.

Attached Deliverables

The following files are attached:

- GNS3 Portable Project ZIP ([Click Here](#))
- Ping test screenshots ([Click Here](#))
- Configure ([Click Here](#))
- Topology ([Click Here](#))
- GitHub ([Click Here](#))